

ECON 712: Applied Econometrics

Lecture 14: Instrumental Variables

Instrumental Variables

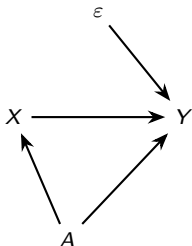
Instrumental Variables and Heterogeneity

Instrumental Variables

- There are instances that we cannot solve issues endogeneity by adding controls, e.g.
 - ▶ Omitted variable bias: the omitted variable is not observed
 - ▶ Treatment Effects: the treatment assignment does not satisfy the CIA assumption for any set of observable covariates
- In this case, we can solve the issue by finding an instrument, Z , that allows us to identify the casual effect of an endogenous variable, X
 - ▶ An instrument is a variable that only affects the outcome variable through X and that does not correlates with unobserved variables affecting the outcome variable

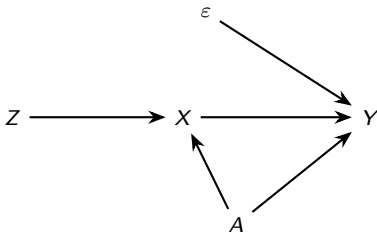
E.g. Omitted variable bias

Suppose the true causal model is:



This gives: $y = \beta_0 + \delta x + \beta_1 A + \epsilon$

- An instrument z satisfies:



E.g. Omitted variable bias

- In this case, if we estimate the relation: $y = \beta_0 + \delta x + \eta$, where $\eta = \beta_2 A + \epsilon$ using a sample and applying OLS to solve for $\hat{\delta}$, we would get a biased and inconsistent estimator
- However, note that: i. $\text{Cov}(z, x) \neq 0$, ii. $\text{Cov}(z, \eta) = 0$ (due to $E[\eta|z] = 0$ —**exclusion restriction**), and iii. z only affects y through x
 - ▶ Question: If we observe z , can we use this variable to help us estimate δ ? The answer is **YES**
- In the case that we do not observe A , but we have a variable z , we could use the following relationship:

$$\delta = \frac{\text{Cov}(y, z)}{\text{Cov}(x, z)} = \frac{\gamma_1}{\alpha_1}$$

where:

$$\begin{array}{lll} y = \gamma_0 + \gamma_1 z + u & E[u|z] = E[u] = 0 & \text{Reduced-Form} \\ x = \alpha_0 + \alpha_1 z + \nu & E[\nu|z] = E[\nu] = 0 & \text{First-Stage} \end{array}$$

- Using a sample to estimate parameters:

$$\hat{\delta}^{IV} = \frac{\hat{\gamma}_1}{\hat{\alpha}_1}, \quad \hat{\gamma}_1 = \frac{\sum_{i=1}^n (y_i - \bar{y})(z_i - \bar{z})}{\sum_{i=1}^n (z_i - \bar{z})^2}, \quad \hat{\alpha}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(z_i - \bar{z})}{\sum_{i=1}^n (z_i - \bar{z})^2}$$

- We cannot show unbiasedness in finite samples, but we can show:

$$plim(\hat{\delta}^{IV} - \delta) = \frac{plim \sum_{i=1}^n (z_i - \bar{z}) \eta}{plim \sum_{i=1}^n (x_i - \bar{x})(z_i - \bar{z})} = \frac{Corr(z, \eta) \sigma_{\eta}}{Corr(x, z) \sigma_x} = 0$$

- In practice, we use a the following procedure called 2SLS (e.g. with one endogenous variable and one instrument)

$$y_i = \mathbf{W}_i^T \beta + \delta \hat{x}_i + \xi_i \quad , \quad E[\xi | \mathbf{W}_i, \hat{x}] = 0$$

- ▶ Where $\hat{x}_i = \mathbf{W}_i^T \pi + \alpha z_i$ and $\xi = \eta_i + \delta \nu_i$
- ▶ Also, $E[\eta | z] = E[\eta]$ (by exclusion restriction), $E[\nu | z] = E[\nu]$ (by first stage)
- That is δ is estimated from the variation of x that is due to z only

- In practice:

1. Estimate π , α from the first stage using OLS and obtain:

$$\hat{x}_i = E[x|z_i] = \mathbf{W}_i^T \hat{\pi} + \hat{\alpha} z_i$$

2. Substitute this into main equation and estimate β and δ using OLS:

$$y_i = \mathbf{W}_i^T \hat{\beta}^{2sls} + \hat{\delta}^{2sls} \hat{x}_i + \hat{\xi}_i \quad , \quad \hat{\xi}_i = \hat{\eta}_i + \hat{\delta}(x_i - \hat{x}_i)$$

- In the case of one endogenous variable and one instrument this estimator is the same as the one presented above:

$$\hat{\delta}^{2sls} = \frac{(\mathbf{M}_w \mathbf{z})^T \mathbf{y}}{(\mathbf{M}_w \mathbf{z})^T (\mathbf{M}_w \mathbf{z})} \frac{1}{\hat{\alpha}} = \hat{\delta}^{IV}$$

- Standard errors for $\hat{\delta}^{2sls}$ are larger than on the OLS case due to:

1. Variation in $\hat{x}$ is a subset of variation in x
2. The uncertainty from the first stage estimation needs to be accounted for

- Suppose you have the following model:

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 w_{1i} + \cdots + \beta_{k+1} w_{ki} + \eta_i$$

- Assume that x is endogenous but that you have j instruments, then:

- ▶ From the first stage you get:

$$\hat{x}_i = \hat{\alpha}_0 + \hat{\alpha}_1 w_{1i} + \cdots + \hat{\alpha}_k w_{ki} + \hat{\alpha}_{k+1} z_{1i} + \cdots + \hat{\alpha}_{k+j} z_{ji}$$

- ▶ The substitute this for x_i into the estimating equation and estimate coefficients:

$$y_i = \hat{\beta}_0 + \hat{\beta}_1 \hat{x}_i + \hat{\beta}_2 w_{1i} + \cdots + \hat{\beta}_{k+1} w_{ki} + \hat{\xi}_i$$

- There could also be multiple endogenous variables as well, in which case we need to write it in vector form

2SLS: General Vector Form

Setup. Partition regressors into exogenous controls $\mathbf{W}$ ($n \times k_1$) and endogenous variables $\mathbf{X}_2$ ($n \times k_2$):

$$\mathbf{y} = \mathbf{W}\alpha + \mathbf{X}_2\delta + \eta \equiv \mathbf{X}\beta + \eta, \quad \mathbf{X} = [\mathbf{W} \ \mathbf{X}_2] \in \mathbb{R}^{n \times (k_1+k_2)}$$

Stack exogenous controls and j excluded instruments into the instrument matrix:

$$\mathbf{Z} = [\mathbf{W} \ \mathbf{Z}_2] \in \mathbb{R}^{n \times (k_1+j)}$$

Order condition (necessary for identification):

- $j = k_2$: just-identified $j > k_2$: overidentified $j < k_2$: underidentified — cannot estimate δ

2SLS estimator. Let $\mathbf{P}_Z = \mathbf{Z}(\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top$:

$$\hat{\beta}^{2SLS} = (\mathbf{X}^\top \mathbf{P}_Z \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{P}_Z \mathbf{y}$$

Intuitively, $\mathbf{P}_Z \mathbf{X}$ purges $\mathbf{X}_2$ of its endogenous component, keeping only the variation attributable to $\mathbf{Z}$.

2SLS: Asymptotic Distribution

We can derive the asymptotic distribution of the **2SLS** estimator as follows:

$$\sqrt{n}(\hat{\beta}^{2sls} - \beta) \xrightarrow{d} N(\mathbf{0}, S_{V^\top V}^{-1} \mathbb{E} [\mathbf{V}_i \eta_i^2 \mathbf{V}_i^\top] S_{V^\top V}^{-1})$$

- where $P_Z \mathbf{X} = V [\mathbf{W} \hat{\mathbf{X}}_2]$

As in the case of OLS:

- For homoscedastic error terms we estimate the variance as:

$$\text{Var}(\hat{\beta}^{2sls}) = \hat{\sigma}_\eta^2 (\mathbf{V}^\top \mathbf{V})^{-1} \quad , \quad \hat{\sigma}_\eta^2 = (n - k)^{-1} \hat{\boldsymbol{\eta}}^\top \hat{\boldsymbol{\eta}}$$

- For heteroscedastic error terms, we use the sandwich estimator:

$$(\mathbf{V}^\top \mathbf{V})^{-1} \mathbf{V}^\top \hat{\boldsymbol{\Sigma}}_\eta \mathbf{V} (\mathbf{V}^\top \mathbf{V})^{-1}$$

2SLS: Asymptotic Distribution

There are **two** important considerations:

1. Residual used in estimating the variance of the error term is:

$$\hat{\eta}_i = y_i - \mathbf{X}^T \hat{\beta}^{2sls}$$

and **not** the residual of the second stage:

$$\hat{\eta}_i + (\mathbf{X}_i - \hat{\mathbf{X}}_i)^T \hat{\beta}^{2sls} = y_i - \hat{\mathbf{X}}^T (\hat{\beta}^{2sls})$$

- ▶ That is implementing 2SLS manually uses the wrong residuals. Use `ivreg` or `ivreg2` in STATA to calculate the correct standard errors (similar command in R)

2. Elements of $(\mathbf{V}^T \mathbf{V})^{-1}$ are hard to analyze, but we know that:

$$\mathbf{V}^T \mathbf{V} = \mathbf{X}^T P_z \mathbf{X}$$

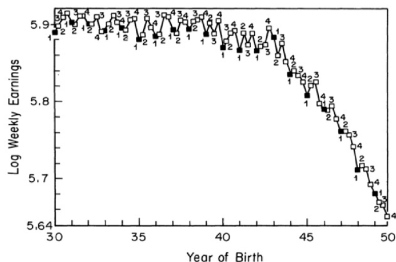
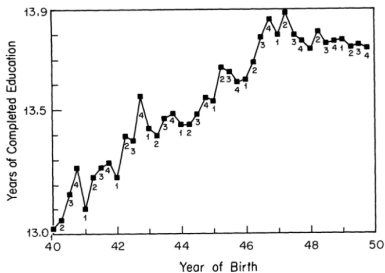
- ▶ Main diagonal are the result of vector multiplication, therefore, if instruments are weakly correlated with the endogenous variables these elements go to zero
- ▶ Weak instruments result in large variances because the first stage is not estimated precisely

2SLS: Further Considerations

1. The problem of **weak** instruments is not only a problem for the SE but it also results in biased 2SLS estimates even in LARGE SAMPLES!!!
 - ▶ To check for weak instruments we run an F-test of excluded instruments on the first stage (with multiple endogenous variables, we run multiple F-Tests) F Test
2. 2SLS only works with linear instruments because the estimated endogenous variable inherits the exclusion restriction property **only** when the first stage is linear
 - ▶ This means that if the endogenous variable is binary, the first stage must be a LPM (running a Logit or Probit first stage is called the **forbidden** regression)
3. The **exclusion restriction** cannot be directly tested as error terms of the main regression are not observed, however, with **more** instruments than endogenous variables a test of overidentifying restrictions can be performed.
 - ▶ This tests for the validity of the instruments included in the IV estimate

IV Example: Angrist and Krueger (1991)

- To resolve issues with endogeneity when measuring the returns to education, Angrist and Krueger (1991) instrumented the years of schooling with the quarter of birth
- The idea is that if students leave school at a particular age, as in over much of the twentieth century, then students whose birthday is in the first quarter will receive less education than students in later quarters:



IV Example: Angrist and Krueger (1991)

VARIABLES	Returns to Education	
	(1) OLS	(2) IV
EDUC	0.062*** (0.000)	0.021 (0.013)
RACE	-0.260*** (0.002)	-0.332*** (0.022)
SMSA	-0.167*** (0.001)	-0.207*** (0.012)
MARRIED	0.251*** (0.002)	0.253*** (0.002)
Constant	4.965*** (0.004)	5.487*** (0.159)
Observations	1,063,634	1,063,634
R-squared	0.293	0.261
Year FE	Yes	Yes
F_test		253.5

Standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Note: The dependent variable is the log weekly earnings and IV variable is *QOB*

IV Example: Angrist and Krueger (1991)

Returns to Education		
VARIABLES	(1) OLS	(2) IV
EDUC	0.062*** (0.000)	0.025** (0.012)
RACE	-0.260*** (0.002)	-0.324*** (0.022)
SMSA	-0.167*** (0.001)	-0.203*** (0.012)
MARRIED	0.251*** (0.002)	0.253*** (0.002)
Constant	4.965*** (0.004)	5.434*** (0.158)
Observations	1,063,634	1,063,634
R-squared	0.293	0.268
Year FE	Yes	Yes
F test		85.21

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Note: The dependent variable is the log weekly earnings and IV variables are dummies for quarter 2, 3,4 of birth

Instrumental Variables

Instrumental Variables and Heterogeneity

Limitations of the Constant-Effects Model

- So far we assumed: $y_i = \alpha + \delta D_i + \eta_i$, where δ is the **same** for everyone
- This rules out that individuals respond differently to treatment — a strong assumption
- In reality, treatment effects are **heterogeneous**:
 - ▶ Education raises wages more for some people than others
 - ▶ Military service affects earnings differently across individuals
 - ▶ Health insurance has different effects on different patients
- With heterogeneous effects, **which** average treatment effect does IV estimate?
- Answer (Imbens & Angrist, 1994): the **Local Average Treatment Effect (LATE)** — the average effect *for the subpopulation whose treatment status is changed by the instrument*

Potential Outcomes Framework

- Each individual i has two **potential outcomes**:

$$\begin{aligned} Y_{1i} & \text{ outcome if treated } (D_i = 1) \\ Y_{0i} & \text{ outcome if not treated } (D_i = 0) \end{aligned}$$

- The **observed** outcome is:

$$Y_i = Y_{0i} + (Y_{1i} - Y_{0i})D_i$$

- The causal **individual treatment effect** is $\delta_i = Y_{1i} - Y_{0i}$:
- Key estimands:

$$\text{ATE} = E[Y_{1i} - Y_{0i}]$$

$$\text{ATT} = E[Y_{1i} - Y_{0i} \mid D_i = 1]$$

$$\text{LATE} = E[Y_{1i} - Y_{0i} \mid \text{complier}] \quad (\text{defined below})$$

- The **fundamental problem of causal inference**: we never observe both Y_{1i} and Y_{0i} for the same person

Introducing the Instrument

- Let $Z_i \in \{0, 1\}$ be a binary instrument that **encourages** treatment but does not determine it
- Define **potential treatment** as a function of the instrument:

D_{1i} treatment status of i if $Z_i = 1$

D_{0i} treatment status of i if $Z_i = 0$

- The **observed** treatment is:

$$D_i = D_{0i} + (D_{1i} - D_{0i})Z_i$$

- Similarly define potential outcomes indexed by both treatment d and instrument z :

$$Y_i(d, z) \quad d \in \{0, 1\}, \quad z \in \{0, 1\}$$

Assumptions about the instrument

1. We assume that $E[\eta_i|Z_i] = E[\eta_i]$, which requires that the following two components are satisfied:

- ▶ **Exclusion restriction** (i.e. the instrument has no direct effect):

$$Y_i(1, 1) = Y_i(1, 0) \equiv Y_{1i}$$

$$Y_i(0, 1) = Y_i(0, 0) \equiv Y_{0i}$$

- ▶ The instrument is **as good as random**:

$$\{Y_i(D_{1i}, 1), Y_i(D_{0i}, 0), D_{0i}, D_{1i}\} \perp\!\!\!\perp Z_i$$

2. **First Stage (Relevance)**:

$$E[D_{1i} - D_{0i}] \neq 0 \iff E[D_i | Z_i = 1] \neq E[D_i | Z_i = 0]$$

3. **Monotonicity (No Defiers)**:

$$D_{1i} \geq D_{0i} \quad \text{for all } i$$

- ▶ The instrument weakly **increases** everyone's probability of treatment — no one is discouraged from treatment by a higher Z_i
- ▶ Rules out **defiers**: individuals who do the opposite of what the instrument encourages

Proposition (LATE Theorem)

Under independence, exclusion restriction, first stage, and monotonicity:

$$\frac{E[Y_i | Z_i = 1] - E[Y_i | Z_i = 0]}{E[D_i | Z_i = 1] - E[D_i | Z_i = 0]} = E[Y_{1i} - Y_{0i} | D_{1i} > D_{0i}]$$

- The left-hand side is the **Wald estimator** — the IV ratio with a binary instrument
- The right-hand side is the **Local Average Treatment Effect**: the average causal effect **for compliers**
- IV does **not** identify ATE or ATT in general — only LATE
- The word “local” means the estimate is local to the complier subpopulation *defined by this instrument* — it changes with the instrument

Step 1 — Reduced Form (Numerator):

$$E[Y_i | Z_i = 1] - E[Y_i | Z_i = 0]$$

By **exclusion restriction**:

$$= E[Y_{0i} + (Y_{1i} - Y_{0i})D_i | Z_i = 1] - E[Y_{0i} + (Y_{1i} - Y_{0i})D_i | Z_i = 0]$$

By **independence** ($Z_i \perp (Y_{0i}, Y_{1i}, D_{0i}, D_{1i})$), this simplifies to:

$$= E[Y_{0i} + (Y_{1i} - Y_{0i})D_{1i}] - E[(Y_{1i} - Y_{0i})(D_{1i} - D_{0i})]$$

By **monotonicity** ($D_{1i} - D_{0i} \geq 0$) and the law of total expectation:

$$= E[\delta_i | D_{1i} > D_{0i}] \cdot P(D_{1i} > D_{0i})$$

Step 2 — First Stage (Denominator):

$$E[D_i | Z_i = 1] - E[D_i | Z_i = 0] \stackrel{\text{indep}}{=} E[D_{1i}] - E[D_{0i}] \stackrel{\text{mono}}{=} P(D_{1i} > D_{0i})$$

Step 3 — Taking the Ratio:

$$\frac{E[Y_i | Z_i = 1] - E[Y_i | Z_i = 0]}{E[D_i | Z_i = 1] - E[D_i | Z_i = 0]} = \frac{E[Y_{1i} - Y_{0i} | D_{1i} > D_{0i}] \cdot P(D_{1i} > D_{0i})}{P(D_{1i} > D_{0i})}$$
$$= E[\delta_i | D_{1i} > D_{0i}] \quad \blacksquare$$

Appendix

F-Test of Excluded Instruments

- To perform an F-test of excluded instruments, we need to run the following two first-stage regressions:

1. Unrestricted:

$$x_i = \hat{\alpha}_0 + \hat{\alpha}_0 w_{1i} + \cdots + \hat{\alpha}_k w_{ki} + \hat{\alpha}_{k+1} z_{1i} + \cdots + \hat{\alpha}_{k+j} z_{ji} + \hat{\epsilon}_i^{ur}$$

2. Restricted:

$$x_i = \hat{\alpha}_0 + \hat{\alpha}_0 w_{1i} + \cdots + \hat{\alpha}_k w_{ki} + \hat{\epsilon}_i^r$$

- From these two regressions, we obtain:

$$SSR^{ur} = \sum_{i=1}^n (\hat{\epsilon}_i^{ur})^2 \quad SSR^r = \sum_{i=1}^n (\hat{\epsilon}_i^r)^2$$

- The F-Test tests the following hypothesis:

$$H_0 : \alpha_{k+1} = \alpha_{k+2} = \cdots = \alpha_{k+j} = 0 \quad H_A = \text{Not all coefficients are zero}$$

F-Test of Excluded Instruments

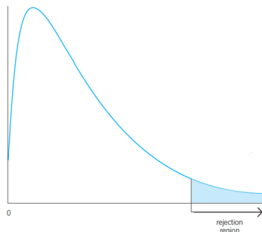
- The F-score is calculated as follows

$$F = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n - k - 1)} \sim F_{q, n-k-1}$$

Number of restrictions

The relative increase of the sum of squared residuals when going from H_1 to H_0 follows a F-distribution (if the null hypothesis H_0 is correct)

- In general, we use the F-Distribution with the appropriate level of significance to test for the Null:



- However, given problems with weak instruments, a rule of thumb is that if $F - score < 10$, then we have a weak instrument problem